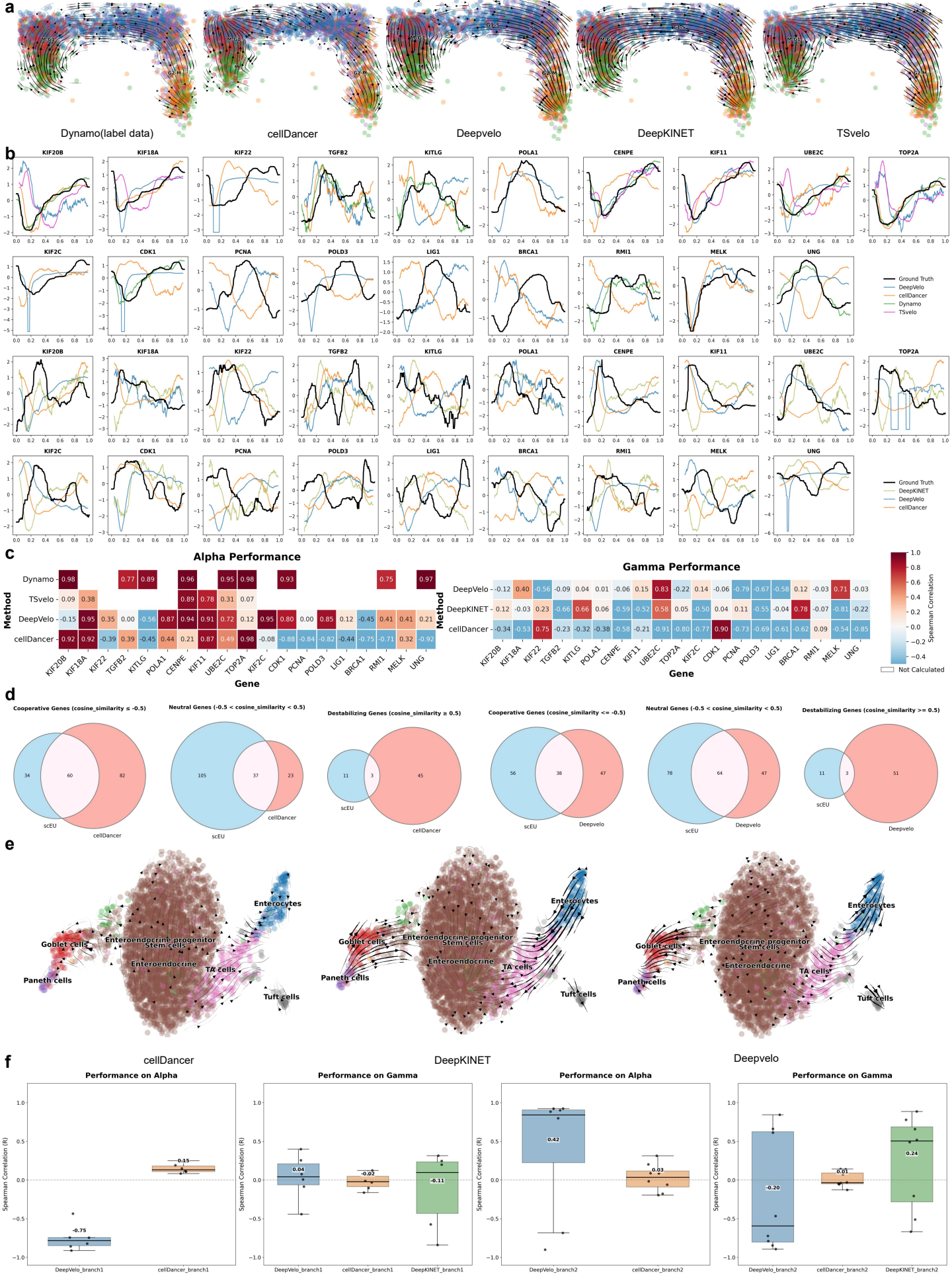

a
Dynamo(label data)
cellDancer
Deepvelo
DeepKINET
TSvelo
b
KIF20B
KIF18A
KIF22
TGFB2
KITLG
POLA1
CENPE
KIF11
UBE2C
TOP2A
KIF2C
CDK1
PCNA
POLD3
LIG1
BRCA1
RMI1
MELK
UNG
Ground Truth
DeepVelo
cellDancer
Dynamo
TSvelo
Ground Truth
DeepKINET
DeepVelo
cellDancer
c
Alpha Performance
Method
Gene
Dynamo
TSvelo
DeepVelo
cellDancer
Gamma Performance
Spearman Correlation
Not Calculated
d
Cooperative Genes (cosine_similarity ≤ -0.5)
Neutral Genes (-0.5 < cosine_similarity < 0.5)
Destabilizing Genes (cosine_similarity ≥ 0.5)
Cooperative Genes (cosine_similarity <= -0.5)
Neutral Genes (-0.5 < cosine_similarity < 0.5)
Destabilizing Genes (cosine_similarity >= 0.5)
scEU
cellDancer
Deepvelo
e
Enterocytes
Goblet cells
Enteroendocrine progenitor
Stem cells
Enteroendocrine
TA cells
Paneth cells
Tuft cells
f
cellDancer
DeepKINET
Deepvelo
Performance on Alpha
Performance on Gamma
Spearman Correlation (R)
DeepVelo_branch1
cellDancer_branch1
DeepKINET_branch1
DeepVelo_branch2
cellDancer_branch2
DeepKINET_branch2